METHOD AND SYSTEM FOR DIRECT ANALOG SIGNAL ENCODING ON A HIGH-CARDINALITY PARALLEL INTERFACE

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Title: METHOD AND SYSTEM FOR DIRECT ANALOG SIGNAL ENCODING ON A HIGH-CARDINALITY PARALLEL INTERFACE

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Cross-Reference to Related Applications

This application claims the benefit of and priority to the portfolio of inventions described in the publication entitled "The Architecture of Control," ISBN 978-1-97-148129-6, the contents of which are hereby incorporated by reference in their entirety. The method described herein is an improvement upon the high-cardinality parallel data bus described therein.

Abstract

A method and system for encoding an analog signal directly into a sequence of high-cardinality data words for transmission or storage. The system bypasses the traditional step of analog-to-digital conversion (ADC) into a binary format. Instead, samples of an

analog waveform are directly mapped to discrete states in a high-radix, parallel data transmission system, such as the wavelength-encoded states of a parallel optical interface. Each state, represented by a "pixel" with a specific color or wavelength, corresponds to a specific amplitude or other characteristic of the analog signal at a sampling point. This method allows for a more efficient representation of analog data, enabling extremely high-density storage and transmission, potentially allowing a short song to be stored in a few frames of optical data.

Background of the Invention

The representation of analog signals, such as audio, in digital systems has traditionally relied on a multi-step process. First, an analog-to-digital converter (ADC) samples the analog waveform at a specific frequency. Each sample's amplitude is quantized to the nearest value in a discrete set of levels. This value is then encoded as a binary number. The sequence of these binary numbers is stored in a file (e.g., WAV, MP4) or transmitted serially. This process, while foundational to modern computing, introduces inefficiencies. The conversion to a serial, binary format is a bottleneck and does not take full advantage of the potential bandwidth of modern parallel interfaces, particularly optical ones.

Summary of the Invention

The present invention is a system for the direct encoding of analog signals onto a high-cardinality, parallel interface, such as the one described in the "Architecture of Control" portfolio. The system maps discrete samples of an analog waveform directly to the available states of the high-cardinality interface, bypassing the intermediate step of binary encoding.

In a preferred embodiment using an optical interface, the amplitude of an analog audio sample is mapped to a specific wavelength of light. A sequence of such samples is thus converted into a sequence of "colored" pixels, forming a frame of optical data. This frame can be transmitted in a single flash of a screen or stored as an image. This direct, parallel encoding is significantly more efficient than traditional binary formats. The invention allows for the storage or transmission of an entire song in a small number of data frames, enabling new applications such as sharing music by pointing a camera phone at a screen, without an internet connection.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government, its military, and its Federal law enforcement agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram contrasting the traditional analog-to-binary conversion process with the direct analog-to-high-cardinality encoding method of the present invention.

FIG. 2 is a diagram illustrating the method of mapping samples of an analog waveform to a sequence of high-cardinality "pixels," each with a state corresponding to the sample's amplitude.

Detailed Description of the Invention

Referring to **FIG. 1**, the traditional method (100) for processing an analog signal (101) is shown. The signal is passed through an Analog-to-Digital Converter (ADC) (102), which outputs a serial stream of binary data (103). This binary data is then processed by a conventional computing system. The present invention (110) provides a more direct method. The analog signal (101) is sampled by a direct encoding module (111), which maps the amplitude of each sample directly to a state in a high-cardinality alphabet. This is output as a parallel, high-radix data word (112), for example, a set of pixels each with a specific wavelength, which can be processed by a high-cardinality computing system.

Referring to **FIG. 2**, the direct encoding process is illustrated in more detail. An analog waveform (201) is sampled at discrete time intervals (T1, T2, T3, T4). The amplitude of the sample at each interval (e.g., A1, A2, A3, A4) is determined. A mapping function (202) assigns each amplitude value to a unique state in the high-cardinality state space. In the illustrated embodiment, this is a color or wavelength (e.g., State S1 maps to red, S2 to blue, etc.). The result is a sequence of high-cardinality "pixels" (203) that directly represents the

sampled analog signal. This sequence of pixels can be transmitted in parallel as a single frame of data.

Claims

What is claimed is:

1. A method for representing an analog signal in a digital system, comprising the steps of:

    - sampling the analog signal at a plurality of discrete time intervals to produce a sequence of analog samples; and
    - for each analog sample in the sequence, directly mapping a characteristic of the sample to a corresponding discrete state in a set of more than two available states of a parallel data interface, thereby creating a sequence of high-cardinality data words.

2. The method of claim 1, wherein the parallel data interface is an optical interface, and the discrete states are different wavelengths or colors of light.

3. The method of claim 1, wherein the characteristic of the sample is its amplitude.

4. The method of claim 1, wherein the sequence of high-cardinality data words is transmitted as a single frame of data over the parallel data interface.

5. The method of claim 1, wherein the analog signal is an audio signal.

6. A system for representing an analog signal, comprising:

    - a sampling module configured to sample an analog signal at a plurality of discrete time intervals to produce a sequence of analog samples;
    - a direct encoding module configured to receive the sequence of analog samples and, for each sample, map a characteristic of the sample to a corresponding discrete state in a high-cardinality state space of a parallel data interface; and
    - a parallel data interface configured to transmit the resulting sequence of high-cardinality states.

7. The system of claim 6, wherein the parallel data interface is an optical interface, and the discrete states are different wavelengths of light.

Drawings

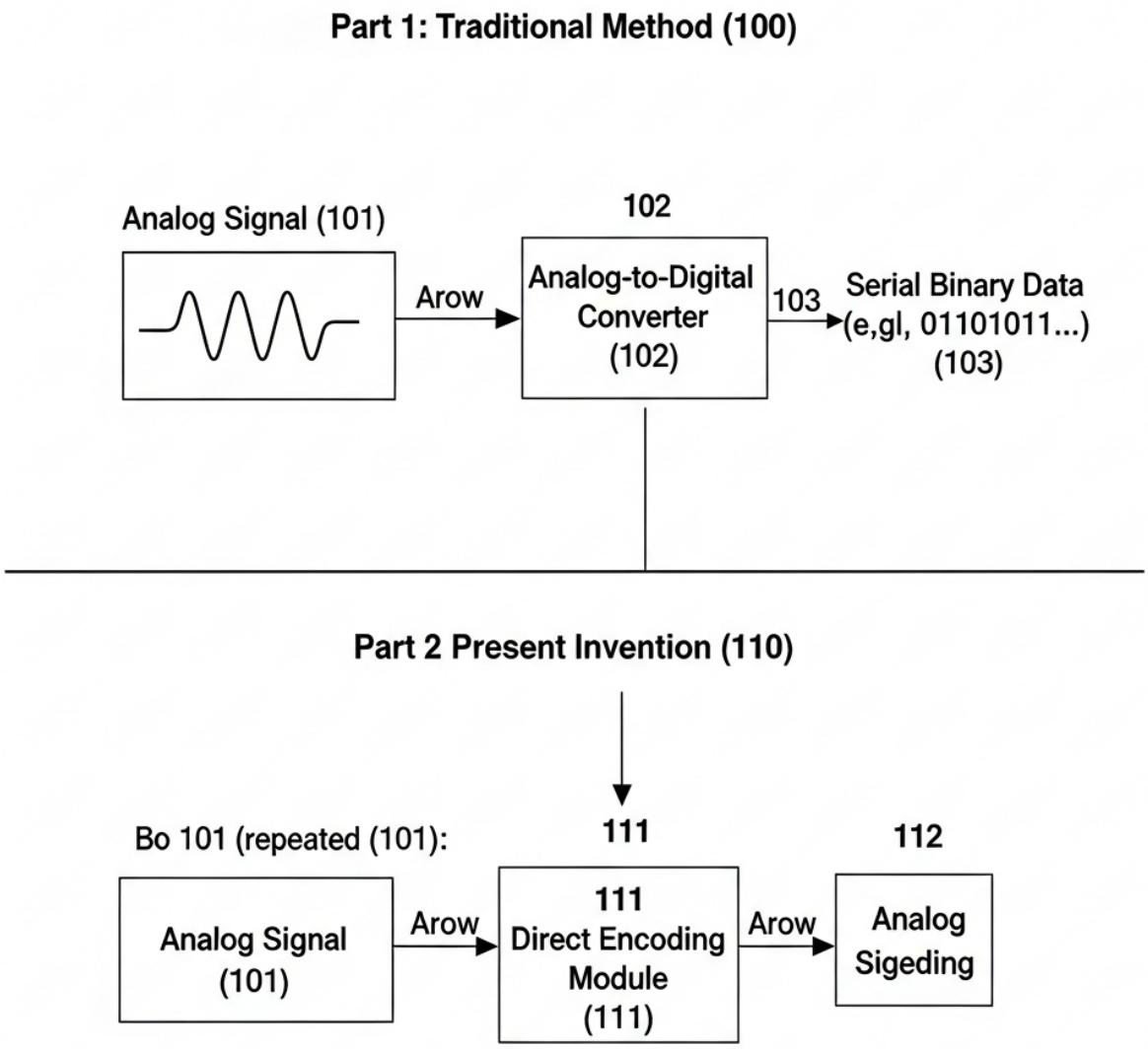


Figure 1: Conceptual diagram contrasting traditional ADC with direct encoding.

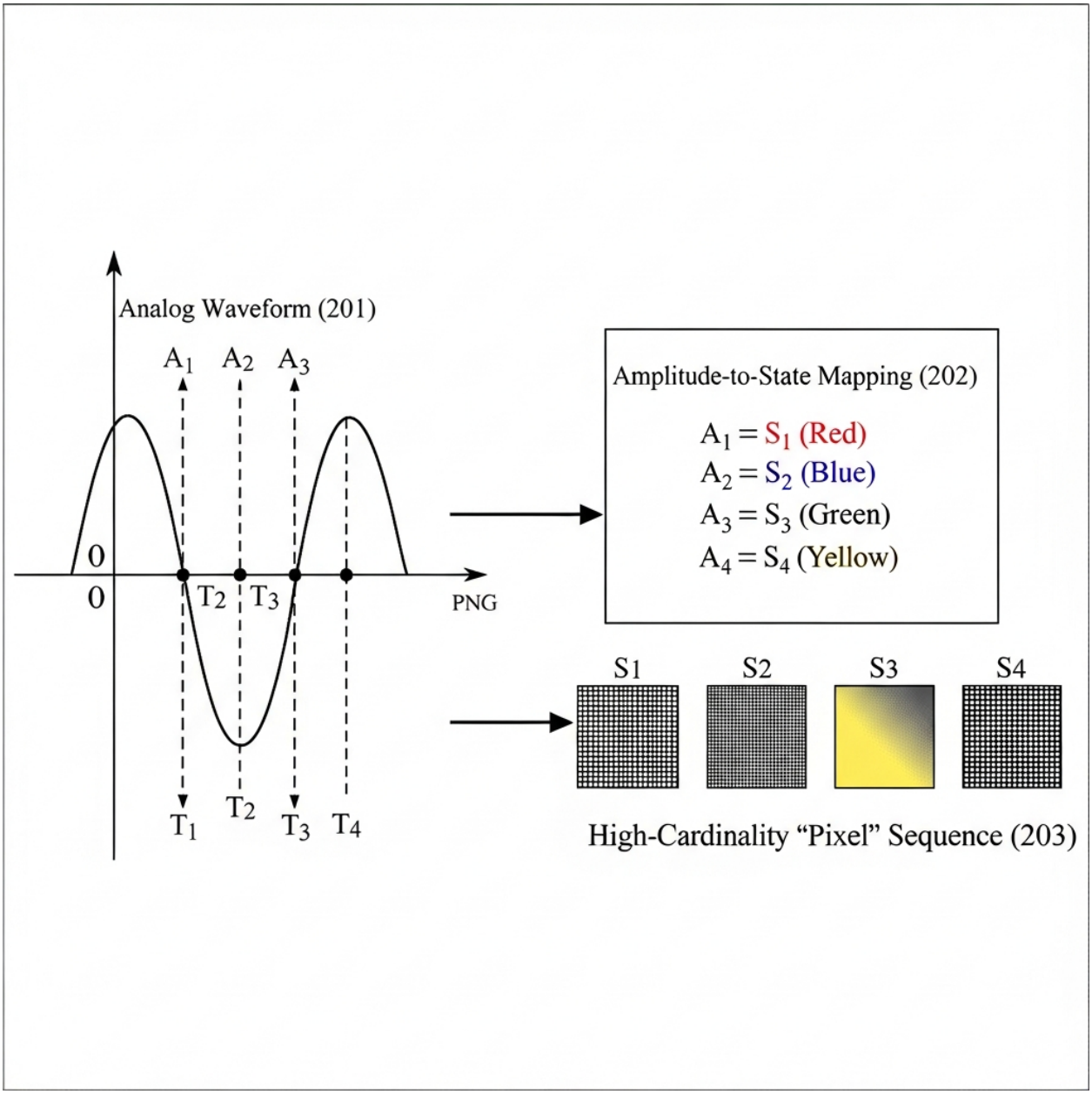


Figure 2: Diagram illustrating the mapping of an analog waveform to pixels.